

1. Executive Summary & System Overview

Sanjeevani AI is an enterprise-grade medical triage assistant, regional multilingual voice bot, and automated e-pharmacy platform engineered specifically for the Indian healthcare ecosystem. The application addresses healthcare accessibility barriers through speech recognition and speech synthesis across all **22 Eighth-Schedule official Indian languages** plus English, an empathetic multi-turn clinical inquiry engine, deterministic red-flag emergency detection, geolocation-based specialist doctor referral, pharmaceutical multi-tablet scheduling, automated 48–72 hour disease recovery follow-ups, and a full-featured Admin CRM with fleet courier rider dispatch logistics.

22 INDIAN LANGUAGES + EN

Web Speech STT & Regional TTS localized across all 22 official scheduled Indian languages with accent fallbacks.

CLINICAL TRIAGE SAFETY

Multi-turn inquiry screening for patient age, current medications, and underlying health conditions (Asthma, Ulcers, BP, Diabetes) before dispensing OTC medications.

E-PHARMACY & FLEET CRM

Multi-item basket, live 5-stage courier tracking timeline, mock UPI/Card gateway, verified rider assignment, and targeted health alerts.

2. High-Level Architectural Flow

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+ | CLIENT LAYER (Browser / Mobile) | | [Voice STT (Web Speech)] [Web Audio Visualizer] [Multilingual
TTS] [Geolocation API] [Admin CRM]| +-----+
+-----+ | HTTP/1.1 REST API / HTTPS JSON v +-----+
+-----+ | APPLICATION SERVER
(Express.js / Node.js) | | * Static File SPA Router (Wildcard fallback for seamless refresh) | | *
Admin Authentication Barrier (x-admin-token Bearer Middleware) | | * Multi-Turn Conversational
Session Cache (pendingTriageInquiries Map) | +-----+
+-----+ | | | v v v +-----+ +-----+
+-----+ +-----+ | CLINICAL TRIAGE ENGINE | | GEODESIC
PROXIMITY | | JSON DATA STORE ENGINE | | * Age & History Screening | | * Haversine Great-Circle
Distance | | * db-store.json | | * Multi-Tablet Sequencing | | * Specialist Referral Mapping | | *
Customers, Orders | | * NSAID/Asthma Safety Gate | | * GPS (Lat/Lng) City Coordinates | | * Fleet
Riders, Audits | +-----+ +-----+ +-----+
+-----+ | v +-----+
+-----+ | INFRASTRUCTURE & DEPLOYMENT LAYER | | * GitHub Version Control & Webhooks *
Render.com PaaS Cloud Runner * render.yaml Blueprint IaC | +-----+
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3. Comprehensive Technology Stack Breakdown

Layer / Domain	Technology / Library	Technical Role & Implementation Details
Runtime Environment	Node.js v18+ LTS / v20	Asynchronous non-blocking runtime executing API endpoints, file-system transactions, and clinical triage rules.
Web Server Framework	Express.js 4.x	REST API routing, static asset serving, CORS handling, body parsing (<code>express.json()</code>), and catch-all SPA routing.
Frontend Architecture	HTML5 Semantic CSS3 Custom Properties Vanilla ES6+ JS	Zero-build framework dependency architecture for ultra-fast load times, zero bundle compilation latency, and seamless browser compatibility.
Voice Processing (STT)	Web Speech Recognition API	Continuous client-side speech-to-text with locale switching across 22 regional Indian languages (e.g. <code>hi-IN</code> , <code>te-IN</code> , <code>ta-IN</code>).
Voice Synthesis (TTS)	Web SpeechSynthesis API	Text-to-speech engine reading out medical advice and dosages in regional dialects with speech rate/pitch modulation.
Audio Visualizer	Web Audio API	Real-time microphone audio spectrum visualizer using <code>AudioContext</code> , <code>AnalyserNode</code> , and HTML5 Canvas.
Geospatial Intelligence	HTML5 Geolocation Haversine Distance Formula	Extracts patient GPS coordinates (Lat/Lng) and calculates geodesic distances (km) to registered specialist hospitals.
Database & Storage	JSON Document Store	Embedded file persistence (<code>data/db-store.json</code>) with in-memory caching, atomic write operations, and safe state serialization.
Security & Auth	Custom Token Middleware Crypto Sanitization	Admin authentication barrier (<code>x-admin-token</code> header), environment password injection (<code>ADMIN_PASS</code>), and client obfuscation.
UI & Icons	FontAwesome 6.4 Free Plus Jakarta Sans Font	Vector iconography and professional clinical typography for clean mobile and desktop views.
Cloud Deployment	Render.com PaaS	Automated containerized web service hosting with automatic SSL/TLS termination and continuous deployment webhooks.
Infrastructure as Code	Render Blueprint (render.yaml)	Declarative infrastructure definition specifying build/start commands, environment variables, and health-check routes.
Automated QA & Test	Node.js ESM Test Suite	End-to-end automated verification script (<code>test-api.mjs</code>) covering multi-turn triage, rider assignments, targeted alerts, and auth tokens.

4. Specialized Clinical Algorithms & Medical Intelligence

MULTI-TURN CLINICAL TRIAGE ENGINE

Unlike naive keyword matchers, Sanjeevani implements a multi-turn clinical inquiry gate (`triageEngine.js`):

- **Age Screening:** Identifies Pediatric (<12), Adult (20–50), or Geriatric (65+) status.
- **Contraindication Guard:** Automatically excludes NSAIDs (Combiflam/Ibuprofen) for patients with Asthma or Gastric Ulcers, safely substituting Paracetamol (Dolo 650).
- **Interactive Quick-Reply Chips:** Emits one-click buttons (`Adult with High BP` , `Asthma History` , etc.) for effortless patient response.
- **Conversational Memory:** Preserves active symptoms across multiple conversation turns without requiring the patient to re-type their issues.

CHRONOLOGICAL MULTI-TABLET SEQUENCING

Medications are scheduled into structured, time-staggered sequences to maximize efficacy and prevent drug clashes:

- **Step 1 (Pre-Meal Antacid):** Pantoprazole / Omeprazole taken 30 minutes before food to prevent gastric acidity.
- **Step 2 (Post-Meal Analgesic):** Paracetamol / Dolo 650 taken strictly after food for fever and body ache.
- **Step 3 (Antihistamines / Night):** Cetirizine 10mg scheduled before bedtime to eliminate daytime drowsiness.
- **Step 4 (Hydration / Continuous):** ORS Electrolyte salts dissolved in water to counter dehydration.

Emergency Red-Flag Escapes: If critical clinical symptoms are detected (e.g. severe chest pain radiating to left arm, acute dyspnea, stroke symptoms, loss of consciousness), the bot completely suppresses OTC medication recommendations and triggers emergency doctor routing with direct telephone dialers and nearest hospital coordinates.

5. Regional Linguistic Engine (22 Scheduled Indian Languages)

Sanjeevani AI provides speech recognition models and speech synthesis voices across all 22 official Eighth-Schedule languages of India, plus Indian English:

Language	Script & Code	BCP-47 Locale	Language	Script & Code	BCP-47 Locale
Hindi (हिन्दी)	Devanagari (hi)	hi-IN	Assamese (অসমীয়া)	Bengali-Assamese (as)	as-IN
Bengali (বাংলা)	Bengali (bn)	bn-IN	Maithili (मैथिली)	Devanagari (mai)	mai-IN
Telugu (తెలుగు)	Telugu (te)	te-IN	Santali (ᱥᱟᱱᱛᱟᱲ)	Ol Chiki (sat)	sat-IN
Marathi (मराठी)	Devanagari (mr)	mr-IN	Kashmiri (كٲشُر)	Perso-Arabic (ks)	ks-IN
Tamil (தமிழ்)	Tamil (ta)	ta-IN	Nepali (नेपाली)	Devanagari (ne)	ne-NP / ne-IN
Urdu (اردو)	Perso-Arabic (ur)	ur-IN	Konkani (कोंकणी)	Devanagari (kok)	kok-IN
Gujarati (ગુજરાતી)	Gujarati (gu)	gu-IN	Sindhi (سنڌي)	Perso-Arabic (sd)	sd-IN
Kannada (ಕನ್ನಡ)	Kannada (kn)	kn-IN	Dogri (डोगरी)	Devanagari (doi)	doi-IN
Odia (ଓଡ଼ିଆ)	Odia (or)	or-IN	Manipuri (মৈতৈলোন)	Meitei (mni)	mni-IN
Malayalam (മലയാളം)	Malayalam (ml)	ml-IN	Bodo (बर')	Devanagari (brx)	brx-IN
Punjabi (ਪੰਜਾਬੀ)	Gurmukhi (pa)	pa-IN	Sanskrit (संस्कृतम्)	Devanagari (sa)	sa-IN
English (India)	Latin (en)	en-IN	Status	22 Languages + EN	Active

6. E-Pharmacy Logistics, Tracking & Courier Fleet CRM

PATIENT ORDERING & TRACKING

- **Multi-Item Pharmaceutical Basket:** Supports itemized quantities, dosage forms, meal guidelines, and subtotal calculation.
- **GPS Delivery Address Autofill:** Uses HTML5 Geolocation to reverse geocode patient coordinates.
- **Simulated Payment Gateway:** UPI (Google Pay, PhonePe, Paytm), Cards, and Cash on Delivery (COD).
- **Multi-Channel Tracking Alerts:** SMS/WhatsApp and Email tracking link dispatch with on-demand resend options.

ADMIN CRM & FLEET ASSIGNMENT

- **Real-time Centralized Order Feed:** View all incoming orders immediately with status filters and customer search.
- **Verified Courier Fleet:** Pre-configured delivery partners with vehicle registrations and direct contact numbers.
- **Custom Courier Dispatch:** Assign custom third-party logistics partners with custom vehicle numbers.
- **Targeted Health Alerts:** Patient selection checklist to dispatch targeted advisories or broadcast alerts.

7. Cloud Deployment & CI/CD Specification

Component	Configuration Specification
Version Control	GitHub(https://github.com/Tharungovindu369/sanjeevani-medical-chatbot.git). Branch: <code>main</code> .
Infrastructure as Code	<code>render.yaml</code> specifying web service <code>sanjeevani-medical-chatbot</code> , <code>env: node</code> , port <code>5000</code> .
Continuous Deployment	Auto-sync on git push to <code>main</code> branch with automated build and container restart.
Health Monitoring	Automated HTTP checks on <code>/api/health</code> returning system uptime and database health.
SSL / TLS	Automatic SSL/TLS termination with Let's Encrypt certificates managed by Render Edge.